

Quiz (Habanero)

- Q1.** What is the vertex of the parabola $y = x^2 + 4x + 4$?
- (A) (0, 4)
(B) (-2, 0)
(C) (1, 5)
(D) (-1, 3)
(E) (-2, 0)
- Q2.** A farmer is modeling the yield of a crop based on temperature. The yield y is related to the temperature x by the equation $y = -x^2 + 20x - 50$. What is the maximum yield?
- (A) 50
(B) 100
(C) 150
(D) 200
(E) 250
- Q3.** What is the axis of symmetry of the parabola $y = x^2 - 6x + 8$?
- (A) $x = 2$
(B) $x = 3$
(C) $x = -2$
(D) $x = -3$
(E) $x = 1$
- Q4.** A city's average temperature is modeled by the equation $y = 2x^2 - 12x + 50$, where x is the month of the year. What is the minimum average temperature?
- (A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q5.** What is the y -intercept of the parabola $y = x^2 - 4x - 3$?
- (A) -3
(B) -2
(C) -1
(D) 0
(E) 1
- Q6.** A researcher is studying the effect of rainfall on plant growth. The growth y is related to the rainfall x by the equation $y = x^2 - 10x + 25$. What is the minimum growth?
- (A) 0
(B) 5
(C) 10
(D) 15
(E) 20
- Q7.** What is the x -intercept of the parabola $y = x^2 + 2x - 6$?
- (A) (-3, 0)
(B) (2, 0)
(C) (-1, 0)
(D) (1, 0)
(E) (3, 0)
- Q8.** A park ranger is modeling the population of a species based on the amount of rainfall. The population y is related to the rainfall x by the equation $y = -2x^2 + 20x - 30$. What is the maximum population?
- (A) 10
(B) 20
(C) 30
(D) 40
(E) 50

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 - 2x - 3$ and identify the vertex, axis of symmetry, and x -intercepts.
- Q10.** A scientist is studying the effect of temperature on the growth of a certain type of algae. The growth y is related to the temperature x by the equation $y = -x^2 + 12x - 20$. Find the maximum growth and the temperature at which it occurs.

Chef's Tips

- Tip 1: Ensure students understand the concept of vertex form
- Tip 2: Have students practice graphing parabolas with different orientations
- Tip 3: Encourage students to use real-world examples to illustrate the applications of quadratic functions